

AI Text-to-Video Website Complete Workflow

This document explains the complete workflow for building an AI-powered text-to-video generation platform that supports:

- AI video generation from text prompts
- Voiceovers and subtitles
- Copyright detection
- Auto SEO generation for YouTube and Instagram
- Secure publishing to social media accounts
- Video downloading and analytics

Phase 1 — Project Planning

Define features, security, APIs, UI design, and SaaS architecture.

Phase 2 — Technology Setup

Set up frontend, backend, database, AI APIs, and storage systems.

Phase 3 — Website Structure

Create frontend and backend folder structures for scalable development.

Phase 4 — User Authentication

Implement JWT login, Google login, OAuth 2.0, and secure sessions.

Phase 5 — AI Script Generation

Generate scripts, hooks, captions, and CTA using AI models.

Phase 6 — Video Generation

Generate scenes, visuals, transitions, and render videos using FFmpeg.

Phase 7 — Voiceover System

Create AI voiceovers using tools like ElevenLabs.

Phase 8 — Subtitle Generator

Generate synchronized captions and animated subtitles.

Phase 9 — Copyright Detection

Detect copyrighted audio/video using fingerprinting and similarity matching.

Phase 10 — Auto SEO Generator

Generate YouTube titles, tags, descriptions, hashtags, and Instagram captions.

Phase 11 — Thumbnail Generator

Create AI-powered thumbnails with text overlays and viral styling.

Phase 12 — Publishing System

Upload videos directly to YouTube and Instagram using OAuth permissions.

Phase 13 — Download System

Allow secure MP4, subtitle, and thumbnail downloads.

Phase 14 — Dashboard & Analytics

Track publishing history, downloads, and engagement analytics.

Phase 15 — Admin Panel

Manage users, API usage, reports, and moderation tools.

Phase 16 — Deployment

Deploy frontend/backend using Vercel, Render, or DigitalOcean.

Phase 17 — Scaling

Use GPU servers, queues, caching, and distributed rendering.

Phase 18 — Legal & Security

Add Privacy Policy, Terms, DMCA, encryption, and secure token storage.

Complete User Workflow

Step	Workflow
1	User logs into the platform
2	User enters a text prompt
3	AI generates script and scenes
4	Visuals and voiceovers are created
5	Subtitles and effects are added
6	Video rendering starts
7	Copyright checking is performed
8	SEO content is auto-generated
9	Thumbnail is created
10	User previews the video
11	User downloads or publishes
12	Analytics and tracking are stored

Recommended Technology Stack

Component	Technology
Frontend	HTML, CSS, JavaScript, React
Backend	Python Flask / FastAPI
Database	MongoDB / PostgreSQL
Video Engine	FFmpeg, MoviePy, OpenCV
Voice AI	ElevenLabs
AI APIs	OpenAI, Stability AI, Runway ML

Storage	AWS S3, Cloudflare R2
Authentication	JWT, OAuth 2.0

Final Product Vision:

“An AI-powered automatic content creation and publishing platform for YouTube, Instagram, Shorts, and Reels with copyright protection, SEO automation, and social media publishing.”